Answer Key - Chemistry - Graduate Level Assessment 13 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** E
   Options: A) 4.5 g; B) 2.2 g; C) 4.0 g; D) 2.0 g; E) 2.5 g
   Note: Correct option: E - 2.5 g

2. **Answer:** A
   Options: A) CaS; B) MgS; C) NaCl; D) LiCl; E) NaF
   Note: Correct option: A - CaS

3. **Answer:** A
   Options: A) 2 mg %; B) 0.0020 mg %; C) 0.002 mg %; D) 0.02 mg %; E) 20 mg %
   Note: Correct option: A - 2 mg %

4. **Answer:** A
   Options: A) 13.00573 amu; B) 13.00128 amu; C) 13.00987 amu; D) 13.00335 amu; E) 13.00739 amu
   Note: Correct option: A - 13.00573 amu

5. **Answer:** A
   Options: A) 135 K; B) 150 K; C) 75 K; D) 110 K; E) 180 K
   Note: Correct option: A - 135 K

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '1'.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '1.61 10^{-23} J/T'.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** False
    Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The wave function and the electron density for a 2 P_Z electron decreases.'.

Long Form Response

11. **Answer:** B_5 pentagons. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 180 Kcal. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key